

Git + RStudio on macOS

Easy setup workflow for students

Goal

At the end, Git is installed, macOS can find it, and RStudio can use it.

Part 1. Install Git

1

Check whether Git is already installed

Open **Terminal** (use Spotlight and search for **Terminal**) and run:

```
git --version
```

If you see a Git version number, Git is installed. Skip to Step 3. If macOS offers to install developer tools, accept the offer and continue to Step 2.

2

Install the Xcode Command Line Tools

You do **not** need to install the full Xcode application. In Terminal, run:

```
xcode-select --install
```

A macOS window should appear. Click **Install**, agree to the license, and **wait until the installation finishes**.

Important: the message `install requested for command line developer tools only` means the installer was requested. It does **not** mean installation is complete.

3

Verify Git, then close and reopen RStudio

After the installer finishes, run:

```
git --version  
which git
```

You should see a Git version number and a path, commonly `/usr/bin/git`. Then **quit RStudio completely** and reopen it.

Part 2. Check that RStudio can find Git

4

Test from the RStudio Console

In the RStudio **Console**, run:

```
Sys.which("git")
```

Good: you get a non-empty path, often `/usr/bin/git`.

Problem: you get only `""`. If that happens, use the troubleshooting page.

5

Check RStudio's Git executable setting

Go to **Tools > Global Options > Git/SVN**. Look at **Git executable**.

For the standard Apple Git installation, it will usually be:

```
/usr/bin/git
```

If you click **Browse** and cannot navigate to `/usr/bin`, press **Command + Shift + G**, type `/usr/bin`, and choose `git`.

6

Restart RStudio, then do a final check

Restart RStudio after changing the Git executable. Then choose **File > New Project....** You should see **Version Control** as an option.

You can also run `Sys.which("git")` again. A non-empty path means R can find Git.

Troubleshooting: Git + RStudio on macOS

Use the first section that matches what you see.

A. git --version keeps asking to install developer tools

1	Make sure you completed the macOS installer window: click Install , agree to the license, and wait until it says the installation is finished.
2	Open System Settings > General > Software Update . If a Command Line Tools update is offered, install it.
3	In Terminal, run <code>xcode-select -p</code> . A normal standalone installation usually reports <code>/Library/Developer/CommandLineTools</code> .
4	After a macOS upgrade, the Command Line Tools may need to be updated or installed again. When finished, quit and reopen RStudio.

B. Terminal finds Git, but RStudio still does not

Set the executable directly in RStudio

Go to **Tools > Global Options > Git/SVN** and set **Git executable** to `/usr/bin/git`. If the Browse window hides that folder, press **Command + Shift + G** and enter `/usr/bin`. Then restart RStudio.

C. The Git tab disappeared after a macOS update

Check the Command Line Tools first

Open Terminal and run `git --version`. A macOS update can require the Command Line Tools to be updated or a license/prompt to be completed again. Follow any specific macOS prompt, then restart RStudio.

D. RStudio has Git, but you do not see a Git tab

The Git tab appears when you are working inside an RStudio Project that is also a Git repository. To test the setup, choose **File > New Project... > New Directory > New Project** and look for the option to create a Git repository, or choose **Version Control** when creating a project.

Fast diagnostic checklist

Check	What you want to see
<code>git --version</code>	A Git version number
<code>which git</code>	A Git path, commonly <code>/usr/bin/git</code>
<code>Sys.which("git")</code>	A non-empty path
RStudio > Tools > Global Options > Git/SVN	<code>/usr/bin/git</code>
RStudio > File > New Project...	A Version Control option

If you are still stuck

Take a screenshot of `git --version`, `which git`, `xcode-select -p`, `Sys.which("git")`, and the RStudio **Git/SVN** settings screen.

Sources

Adapted from *Happy Git and GitHub for the useR*, especially Chapter 6 (Install Git), Chapter 13 (Detect Git from RStudio), and Chapter 14 (Troubleshooting): <https://happygitwithr.com/install-git>, <https://happygitwithr.com/rstudio-see-git>, and <https://happygitwithr.com/troubleshooting>. Installation details also checked against Apple Developer documentation: <https://developer.apple.com/documentation/xcode/installing-the-command-line-tools/>.